O-D-00234a(GSA-FSS)

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Interim Revision of
Fed. Spec. O-D-234
March 22, 1965

INTERIM FEDERAL SPECIFICATION

DEVELOPER, PHOTOGRAPHIC, HIGH CONTRAST

This Interim Federal Specification was developed by General Services Administration, Federal Supply Service, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification O-D-234, dated March 22, 1965.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the performance requirements for a high contrast developer, composed of mixed chemicals packaged in powder form, for use in processing black and white photographic film to high contrast.

1.2 Classification. The developer shall be furnished in the following sizes containing sufficient powder to prepare the quantities of working solution listed below:

Size	Working Solution
1	1 gallon or 3.785 liters
2	2 gallons or 7.571 liters
3	3.5 gallons or 13.248 liters
4	5 gallons or 18.927 liters

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids, or request for proposal, form a part of the specification to the extent specified herein.

Federal Specifications:
- L-F-330 - Film, Photographic, Black and White, General Specification for
- O-C-275 - Chemicals, Photographic "Single Substance", General Specification for
- PPP-B-636 - Box, Fiberboard
- PPP-C-96 - Cans, Metal, 28 Gauge or Lighter
- PPP-P-16 - Packaging of Photographic Dry Chemicals, General Specification

Federal Standards:
- Fed. Std. No. 102 - Preservation, Packaging, and Packing Levels
- Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies)
- Fed. Std. No. 170 - Film, Photographic, Black and White, Classification and Testing Methods

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standards:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

United States of America Standards Institute Standards:
USAS PH4.1 - Manual Processing of Black-and-White Photographic Films and Plates, Methods for
USAS PH4.125 - para-Methylaminophenol Sulfate
USAS PH4.126 - Hydroquinone
USAS PH4.200 - Potassium Bromide
USAS PH4.227 - Sodium Carbonate, Monohydrate
USAS PH4.275 - Sodium Sulfite, Anhydrous
(Copies of the USA Standards may be obtained from the United States of America Standards Institute; 10 East 40th Street, New York, New York 10016).

3. REQUIREMENTS

3.1 Materials. The photographic developer procured in accordance with this specification shall meet all the requirements specified herein. The materials used shall be freely soluble in water and relatively nontoxic.

3.2 Data. No data is required by this specification or by applicable documents referenced in section 2, unless specified in the contract or order (see 6.2).

3.3 Chemicals. The chemicals used shall be photographic grade conforming to O-C-275 or the appropriate USA standard as applicable. If neither a Government specification nor a USA standard is applicable, then a grade of chemical sufficiently pure for photographic use shall be selected.

3.3.1 Formula. No formula is specified for the powder or the liquid developer. This specification calls for either a single-powder preparation or a two-powder preparation.

3.4 Performance characteristics. The performance of the developer when prepared according to the instructions for use on the label shall be equal to, or better than, the performance of a freshly prepared solution mixed according to the formula specified in table I. Testing shall be done in accordance with 4.4.

TABLE I

Test Formula - To Make One Liter

Chemical Name		
para-Methylaminophenol Sulfate	1.0	gram
Sodium Sulfite, Dessicated	75.0	grams
Hydroquinone	9.0	grams
Sodium Carbonate, Monohydrated	30.0	grams
Potassium Bromide	5.0	grams
Sodium Hexametaphosphate	0.75	gram
Water	1.0	liter

3.5 Physical properties. The dry chemicals used in the developer powders shall be suitably pulverized or granulated to eliminate lumps and insure a free flowing mixture in the powder form. The chemicals shall be thoroughly blended so the powder in each container is a homogeneous mixture.

3.5.1 Color of powder. The dry developer powder shall be light in color and free from brown coloration, dark specks, or other evidence of deterioration.

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3.5.2 Solubility and appearance. The developer powder shall be readily soluble when dissolved according to the directions marked on the package. There shall be no dirt or insoluble precipitates when the solution is cooled to 68° F. The color and clarity shall be comparable to that of a freshly prepared solution mixed according to the test formula specified in table I. There shall be no precipitation when the developer solution is cooled to 60° F.

3.6 Sensitometric characteristics. The sensitometric performance of the developer when prepared according to the instructions for use on the label, shall match the performance of a freshly prepared developer, mixed according to the test formula specified in table I, in passing the photographic test in 4.4.2.

3.7 Storage life. The developer powder, in the unit package, shall meet storage temperature ranges of plus 120° to minus 65° F. and shall pass the accelerated aging test and the photographic test given in section 4.

3.8 Workmanship. Workmanship shall be in accordance with high grade commercial practice covering the manufacture and packaging of photographic chemicals.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.1.1 Materials and components. The supplier is responsible for insuring that materials and components were manufactured, tested, and inspected in accordance with the requirements of referenced specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.2 Quality conformance inspection sampling. Unless otherwise specified, sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105. The manufacturer's sampling and inspection procedures may be utilized provided such sampling and inspection methods will assure quality equal to or better than that obtained by the following provisions set forth in MIL-STD-105, and provided the manufacturer's methods have been approved by the procuring activity.

4.2.1 Inspection lot. For purposes of sampling, an inspection lot shall consist of all material of the same classification designator submitted for inspection and delivery at one time.

4.2.1.1 Unit package. For the purpose of sampling the completed unit packages, a lot shall consist of all the unit packages offered for delivery at one time.

4.2.1.2 Developer powder. For the purpose of sampling the developer powder, a lot shall consist of all the chemicals packaged from the same batch and offered for delivery at one time. A batch shall consist of that quantity of chemicals prepared as one integral mix.

4.3 Quality conformance inspection. The examination of the unit package shall be made in accordance with the classification of defects, inspection levels and acceptance levels (AQL's) set forth below.

4.3.1 Appearance, workmanship, and marking. Examine the unit package for defects in appearance, construction, workmanship, closure, and markings. For determining sample size, the inspection level shall be I and the AQL shall be 1.5.

Examine	Defects
Appearance, construction, and workmanship	Cracked, crushed, or any defect affecting serviceability.
Closure	Any leakage or seepage of contents. Closure not as specified.
Markings	Label data omitted, illegible, incorrect, incomplete or not in accordance with requirements.

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4.3.2 Closure. Examine the closure of the unit package to determine that it is waterproof and airtight. For determining sample size, the inspection level shall be S-2 and the AQL shall be 4.0. Samples used for this examination may be used for subsequent examinations.

Examine	Defects
Conduct the leakage test specified in PPP-C-96	Leakage indicated by escaping air

4.3.3 Inner container. Examine the unit package contents for the appearance of the powder, inside of the outer container, construction of the closure, workmanship of the inner container, and net weight. For determining sample size, the inspection level shall be S-2 and the AQL shall be 4.0.

Examine	Defects
Space utilization of container(s) Not required if waived	Less than 80 percent of internal space utilized.
Appearance of the powder(s)	Not in powder form, not homogeneous. Contains lumps, not free flowing. Presence of foreign matter. Any discoloration or other evidence of deterioration. Any leakage or seepage of contents of inner container.
*Appearance, construction, and workmanship of inner container	Cracked, crushed, or any defects affecting serviceability. Not in container specified. Not clean.
Closure of inner container	Any leakage or seepage of contents. Closures not as specified.
Marking of inner container	Label or marking omitted, illegible, incorrect or incomplete.
**Inside coating of outer container	Missing, does not completely cover inner surfaces, flaked or chipped.

* Contractor shall furnish a Certificate of Compliance that inner container is constructed of inert materials or has been coated both inside and outside with a material that will not react either physically or chemically with the contents of the containers.

** Contractor shall furnish a Certificate of Compliance that inner coating will not react physically or chemically with the contents of the containers.

4.4 Test methods. Each batch of developer powder shall be subjected to the following tests. The sample unit shall consist of two unit packages selected at random from the lot. Failure to pass one or more tests shall be cause for rejection of the lot.

4.4.1 Solubility test. Verify the free flowing characteristics of the developer powder from one of the sample unit packages. Prepare a developer solution, using distilled water, from this powder according to the directions for mixing on the container. The developer powder shall dissolve readily without excessive stirring. Prepare a control solution, using distilled water, according to the test formula specified in table I. Components of the control solution shall be weighed with an accuracy of plus or minus one percent. Cool both solutions to 68° F. and examine the test solution for clarity, dirt, and insoluble precipitates. Compare the color of the test solution with the color of the freshly mixed control solution. Cool the test solution to 60° F. There shall be no precipitation of any of the chemicals. The solutions prepared for this test may be used for subsequent tests.

4.4.2 Photographic test. Expose six sheets of high contract, process type, panchromatic film, as specified in L-F-330, of the same emulsion number in an intensity scale sensitometer. Stabilize the solutions prepared for the solubility test and all other processing solutions to a temperature of 68° F. plus or minus one-half degree. Using the processing procedures described in USAS PH4.1, process three sheets of exposed film in the control solution for four minutes and three sheets in the test developer solution for four minutes. After drying, the densities of all the sheets of processed film shall be measured with a densitometer, in accordance with Fed. Std. 170. The densities of each set shall be averaged and a sensitometric curve plotted for each set. To pass the photographic test, the curve from the test developer solution shall have a base plus fog

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density not exceeding 0.04 density unit and shall have a gamma not less than the curve from the control. The film processed in the test developer solution shall show no evidence of mottling, uneven densities, or any other developing defects.

4.4.3 Accelerated aging test. Two unit packages shall be selected from each batch lot and stored, undisturbed, at 120° F. plus or minus three degrees for a period of four weeks. The samples shall then be cooled to room temperature and one sample unit package shall be examined. The developer powder shall not be dark in color, shall be free of dark specks, shall display no evidence of lumpiness or caking, shall not have been contaminated by the container(s), and shall readily dissolve. There shall be no corrosion of the container(s). The test developer made from the stored powder shall pass the photographic test. In the photographic test, the time required for the test developer to develop the exposed film to a gamma of 4.0 plus or minus 0.25 shall not be more than two minutes longer than the time required for the control solution to develop the film to the same gamma. Failure of the sample unit package to pass the photographic test shall be cause for rejection of the entire batch. If the unit package fails to pass the other portions of the storage life test, the second unit shall be examined. Failure of the second unit package to pass this test shall be cause for rejection of the entire batch.

4.5 Test data. Unless otherwise specified in the contract or order (see 6.2), the contractor shall keep the test data specified herein, and a copy shall be made available to the Government inspector.

a. Test data shall be kept on each of the individual bulk chemicals used in the developer to insure that each chemical meets the chemical quality requirements. In lieu of such data, the contractor shall keep on file a certification and an average analysis from the manufacturer of each bulk chemical to prove that the chemicals supplied were in accordance with this specification.

b. Test data shall be kept on each batch of developer powders to assure that it conforms to the performance requirements specified herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, as defined in Fed. Std. No. 102.

5.1.1 Intermediate package. The developer powder unit packages shall be overpackaged in intermediate containers conforming to PPP-B-636, type I, class 1, of any style. The intermediate container shall hold no more than 24 one-gallon or 12 two-gallon or three-one-half gallon or five-gallon sizes of unit packages, as applicable.

5.1.1.1 Single powder package. The unit package shall conform to the requirements of PPP-P-16, type II, class 1 and PPP-C-96, type I, round metal cans. The can may be provided with a rip or scored tab for opening the can with the fingers.

5.1.1.2 Two powder package. The unit package shall consist of two containers referred to as inner and outer container.

5.1.1.3 Outer container. The outer container shall conform to PPP-P-16, type II, class 1, round, metal can and shall be provided with a key-wind, rip-type opening device and key.

5.1.1.4 Inner container. The inner container may be constructed of any suitable material. The material used shall be inert with respect to the chemicals used in the developer, or shall be coated both inside and outside with a suitable material which is inert to the chemicals and the container. The inner container shall be so constructed and closed that the chemicals therein will not come in contact with the chemicals in the outer container. The inner container shall be suitably marked to indicate that its contents will be dissolved first.

5.1.2 Dimensions. Dimensions of the containers shall be such that not less than 80 percent of the internal volume will be utilized and such that the contents will not spill when the containers are opened in an upright position. If regularly available commercial sizes, meeting this requirement, are not available, the requirements may be waived by the requiring agency if requested, with affidavits by the supplier.

5.1.3 Closure. The completed unit package shall be waterproof and air-tight.

5.1.4 Labeling. Each unit package shall be durably and legibly labeled by lithographic or other suitable means. If gummed labels are used they shall be the wrap-around type. The labels shall be affixed in place with water resistant adhesive. The adhesive shall not smear or blur the markings. The following information shall be on the label.

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a. Item identification.
b. Directions for mixing and use.
c. Manufacturer's name and code number.
d. Package size.
e. Net weight.
f. Date of manufacture.

5.2 Packing.

5.2.1 Level A. Intermediate packages of developer powder shall be packed for shipment according to the requirements of PPP-P-16, type I, class 1, of any style. The intermediate containers shall hold no more than 24 one-gallon or 12 two-gallon or three and one-half-gallon or five-gallon sizes of unit packages, as applicable.

5.2.2 Level C. Intermediate packages of developer powder shall be packed for shipment in exterior type shipping containers in a manner that will insure safe transportation at the lowest transportable cost to the point of delivery. Containers shall meet Interstate Commerce Commission Regulations and regulations of common carriers as applicable to the mode of transportation.

5.3 Marking and labeling.

5.3.1 Container marking. Unless otherwise specified by the procuring activity (see 6.2) interior packages and shipping containers shall be durably marked and labeled in accordance with Fed. Std. No. 123, for civilian activities, and MIL-STD-129 for military activities. The marking shall include the Federal Stock Number and the manufacturer's batch number.

6. NOTES

6.1 Intended use. The developer powder specified is intended for use in preparing the working solution of a general developer for producing photographic negatives of high contrast.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification
(b) Stock number
(c) Size of unit package desired (see 1.2)
(d) Quantity desired
(e) Responsibility for inspection (see 4.1)
(f) Selection of applicable levels of packing (see 5.2.1 and 5.2.2)
(g) Marking and labeling (see 5.3.1)
(h) Test data, when required (see 3.2 and 4.5)
(i) Can opener when required.

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